少量データで不確かさも示す(見本)

Report uncertainty with small data (sample)

合成データによる見本。研究結果ではない

Synthetic-data sample, not research results

本発表の要点

Key points

・n = 20 で GPR の誤差が最小

GPR has the smallest error at n = 20

・95% 区間に 100% の点が入る

The 95% interval covers 100% of points

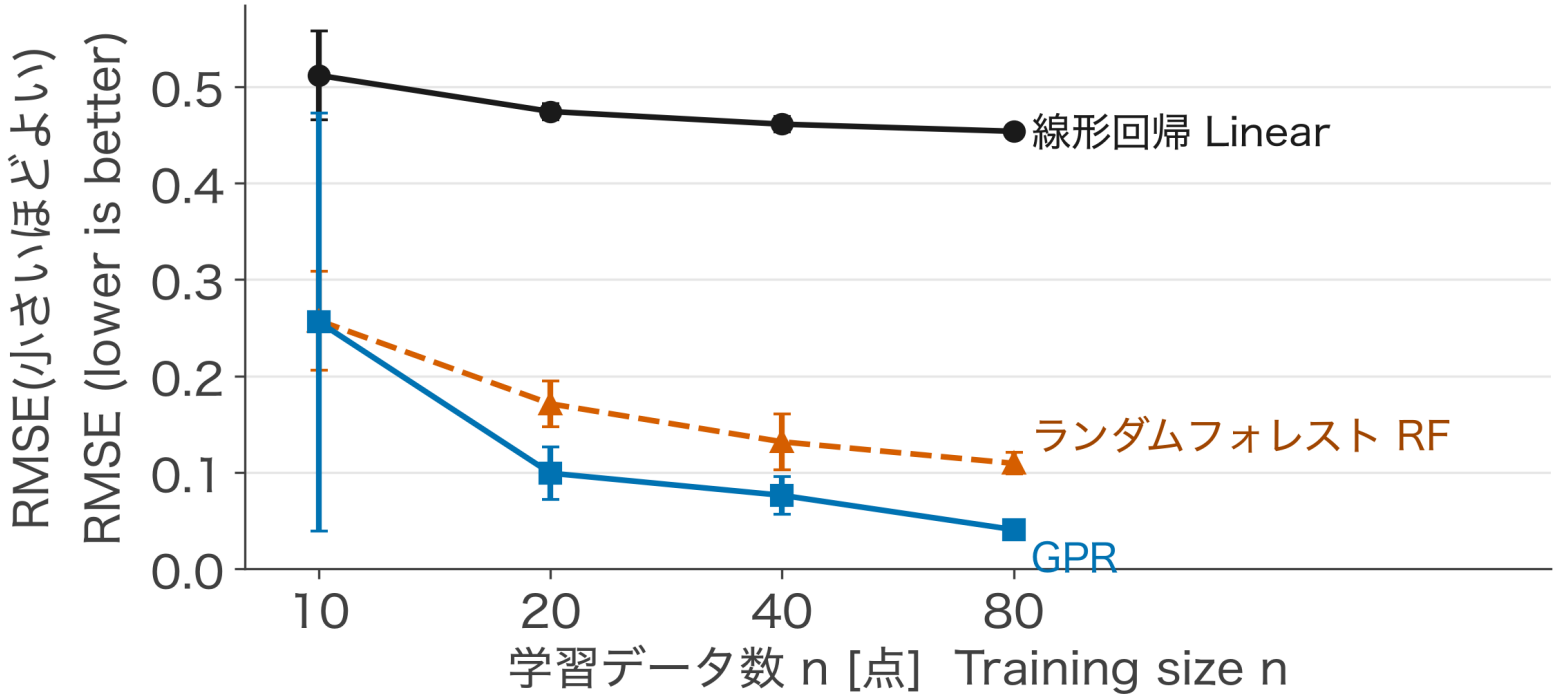
発表者 氏名　所属・学会名・発表日

少量データでは GPR の誤差が最小

GPR has the smallest error with small data

予測誤差 RMSE を 3 手法で比べた

Prediction error (RMSE) of three methods



GPR の RMSE(n = 20)

GPR RMSE at n = 20

0.091

線形回帰は 0.490

Linear: 0.490

GPR = ガウス過程回帰。見本データ、誤差棒 ±1 σ

GPR: Gaussian process regression. Sample data; bars ±1 σ

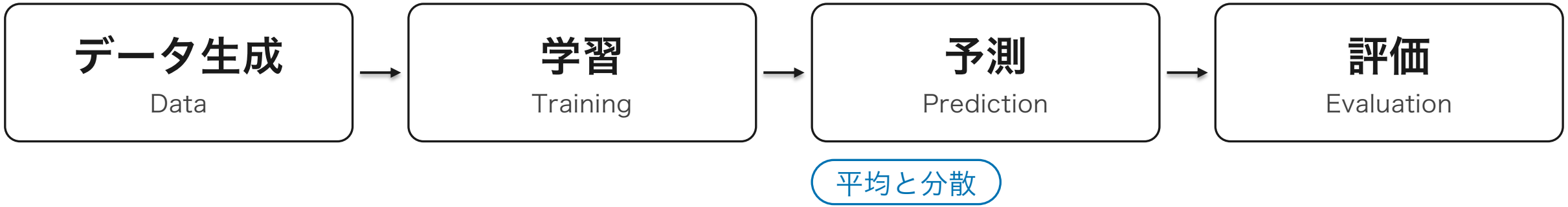
[1] Rasmussen & Williams, MIT Press 2006 [2] Breiman, Mach. Learn. 2001

GPR は予測の平均と分散を同時に返す

GPR returns both the mean and the variance

手法の流れと、予測に使う 2 つの式

Workflow and the two predictive equations



$$\bar{f}_* = \mathbf{k}_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y} \tag{1}$$

$$\mathbb{V}[f_*] = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{k}_* \tag{2}$$

(1) 予測の平均、(2) 予測の分散(不確かさ)[1]。記号は p.4

(1) predictive mean; (2) predictive variance (uncertainty) [1]. Symbols on p.4

[1] Rasmussen & Williams, MIT Press 2006

式 (1)(2) の記号と計算条件

Symbols in Eqs. (1)(2) and conditions

p.3 の式で使う記号の意味

Meaning of the symbols on p.3

記号 Symbol	意味 Meaning
x^*	予測したい点の記述子 Descriptor of the query point
k^*	学習点と x^* の共分散 Covariance between training points and x^*
K	学習点どうしの共分散行列 Covariance matrix of training points
σ_n^2	観測雑音の分散 Noise variance

条件

Conditions

テスト 200 点

200 test points

乱数 5 回

5 random seeds

雑音 $\sigma = 0.15$

Noise $\sigma = 0.15$

[1] Rasmussen & Williams, MIT Press 2006 [3] Pedregosa et al., JMLR 2011

GPR は学習点のない端で σ が大きい

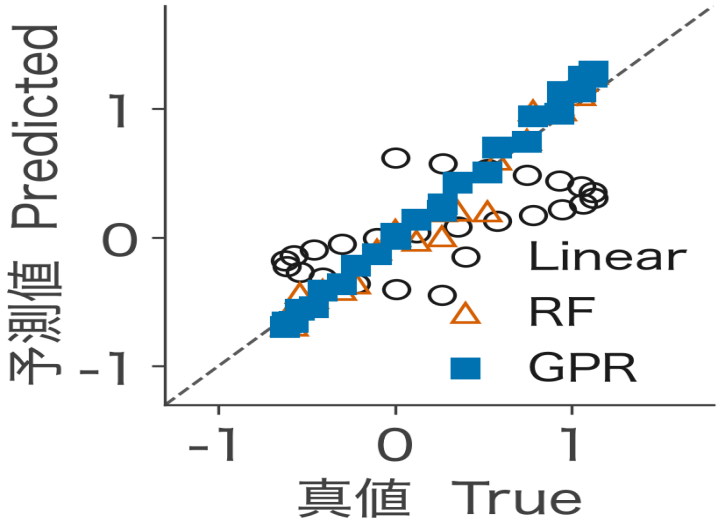
GPR's σ grows where training points are absent

n = 20・乱数 0 の代表例。同じ予測から作図

Representative case (n = 20, seed 0) from the same prediction

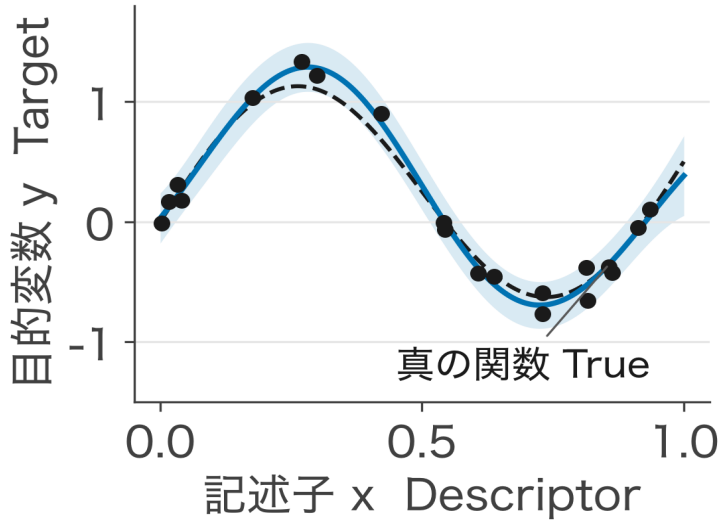
(a) 予測値と真値

(a) Predicted vs. true



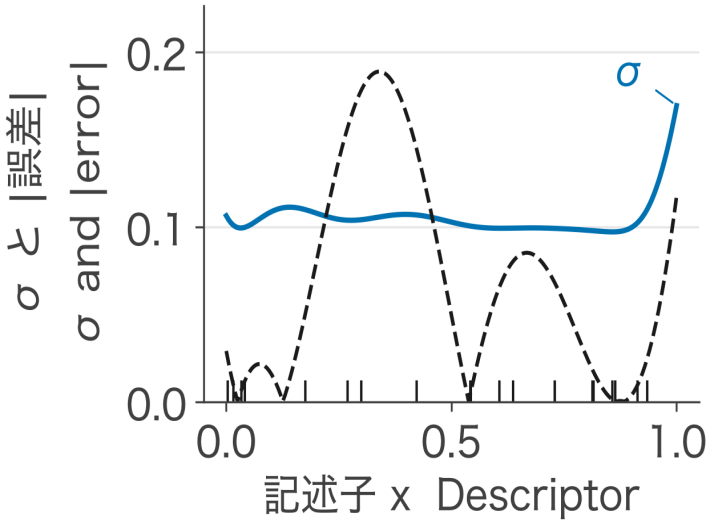
(b) 平均と 95% 区間

(b) Mean and 95% interval



(c) σ と |誤差|

(c) σ and |error|



見本データ。(b) の帯は平均 ± 1.96σ、(c) の下端の縦線は学習点

Sample data. Band in (b): mean ± 1.96σ; ticks in (c): training points

[1] Rasmussen & Williams, MIT Press 2006 [3] Pedregosa et al., JMLR 2011

データが増えると手法の差は縮む

The gap between methods closes with more data

テスト RMSE の平均(下段は ±1 σ、5 回)。小さいほどよい

Mean test RMSE (±1 σ below, 5 seeds). Lower is better

手法 Method	n = 10	n = 20	n = 40	n = 80
線形回帰 Linear	0.512 ±0.046	0.474 ±0.008	0.461 ±0.008	0.454 ±0.005
ランダムフォレスト RF	0.257 ±0.052	0.171 ±0.024	0.132 ±0.029	0.109 ±0.011
GPR	0.256 ±0.217	**0.099** ±0.027	0.076 ±0.019	0.041 ±0.009

読み方: n = 20 では GPR が最小、n = 80 では RF との差が小さい

At n = 20 GPR is lowest; at n = 80 the gap to RF is small

[1] Rasmussen & Williams, MIT Press 2006 [2] Breiman, Mach. Learn. 2001 [3] Pedregosa et al., JMLR 2011

少量データでは不確かさも示す

With small data, report uncertainty too

p.2–6 の結論と限界

Conclusions and limitation of pp. 2–6

結論

Conclusions

・誤差は線形回帰より 82% 小さい (p.2)

Error is 82% below linear (p.2)

・データが増えると差は縮む(p.6)

The gap closes with more data (p.6)

・95% 区間に 100% の点が入る

The 95% interval covers 100% of points

GPR の RMSE(n = 20)

GPR RMSE at n = 20

0.091

線形回帰は 0.490

Linear: 0.490

限界 1 次元の例だけ

Limitation: 1-D example only

[1] Rasmussen & Williams, MIT Press 2006 [2] Breiman, Mach. Learn. 2001 [3] Pedregosa et al., JMLR 2011

参考文献

References

本文の登場順。DOI と Web 資料の確認日つき

In order of citation, with DOIs and access dates

[1] Rasmussen, C. E.; Williams, C. K. I. Gaussian Processes for Machine Learning; MIT Press: Cambridge, MA, 2006. https://gaussianprocess.org/gpml/ (accessed 2026-10-20).

[2] Breiman, L. Random Forests. Mach. Learn. 2001, 45 (1), 5–32. https://doi.org/10.1023/A:1010933404324

[3] Pedregosa, F.; Varoquaux, G.; Gramfort, A.; et al. Scikit-learn: Machine Learning in Python. J. Mach. Learn. Res. 2011, 12, 2825–2830.

[4] Okabe, M.; Ito, K. Color Universal Design (CUD): How to Make Figures and Presentations That Are Friendly to Colorblind People. J*Fly, 2002 (rev. 2008). https://jfly.uni-koeln.de/color/ (accessed 2026-10-05).

[5] Wong, B. Points of View: Color Blindness. Nat. Methods 2011, 8 (6), 441. https://doi.org/10.1038/nmeth.1618